College of Dentistry
3rd Stage
Prosthodontics II
2025-2026
Dr Azal Hadi
Lecture 2

Steps in the Fabrication of Removable Partial Denture

Learning objectives

- Understand the key stages involved in the fabrication of a removable partial denture and their sequence.
- Differentiate between the fabrication process of a removable partial denture and a complete denture.
- Identify and explain the importance of pre-prosthetic procedures in preparing the patient's oral cavity for denture fabrication.
- Describe the process of designing and fabricating the metal framework of a removable partial denture.

Introduction:

Removable partial denture fabrication involves several stages, which are more complex than those for complete dentures, with a few additional procedures..

1. **Diagnosis**

Diagnosis is the determination of the *prosthetic necessity* of the patient. Diagnosis comprises various procedures that are used to evaluate certain conditions in the oral cavity. The diagnostic procedures can generally be grouped into personality evaluation, clinical and laboratory evaluation.

Alternatives to RPDs (Treatment Options):

A. No Treatment (Shortened Dental Arch) - Most patients can function with a shortened dental arch.


photo by (Singh et al , 2016)

B. Fixed partial denture – requires abutments at opposite ends of the edentulous space, more expensive than RPD, must grind down abutments, flexes and can fail if too long.

C. Implant-supported prosthesis - most costly, closest replacement to natural dentition, less costly over the long term


photo by (Mijiritsky et al, 2015)

D. Complete denture; if few teeth are left, with poor prognosis and if the replacement of missing teeth is very complex or costly.

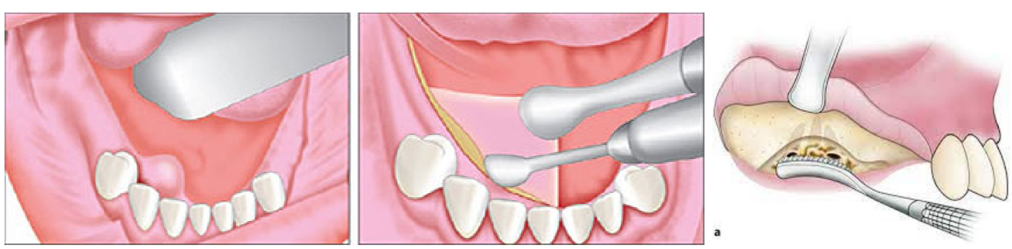
2. **Treatment Planning**

Based on a thorough evaluation of diagnostic data and the patient's oral condition, a customised treatment plan is developed. During this process, the dentist forms a clear idea of the most suitable denture type for the patient.

The treatment will contain details about the pre-prosthetic procedures required for the patient, the technique of impression making, the equipment preferred to carry out various recording procedures, the material to be used for preparing the framework and the denture base, etc.

3. **Pre-prosthetic procedures**

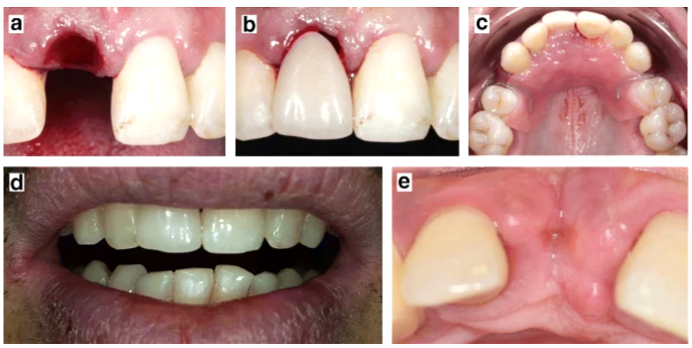
Pre-prosthetic procedures may include periodontal preparation, tissue conditioning, abutment teeth preparation, and pre-prosthetic oral surgical preparation.



4. **Making the Primary Impression and Cast**

Primary impression is taken after all pre-prosthetic procedures have been undertaken (except in the case of an immediate denture, where the primary impression is taken before extraction). Alginate is chosen because it is economical, elastic, and easy to manipulate. The alginate impression is poured using dental plaster. The resulting primary cast is used to design the prosthesis.

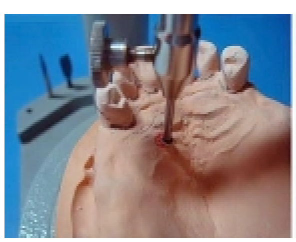

photo by dentulous32wmt



Immediate denture by (Virard et al, 2018)

5. Designing the Prosthesis

Designing includes choosing the type of components for the partial denture, determining the location of various components, determining the path of insertion and choosing the type of material for each component, etc. Designing can be done using an instrument called Surveyor.

Surveying→   ←designing of prosthesis

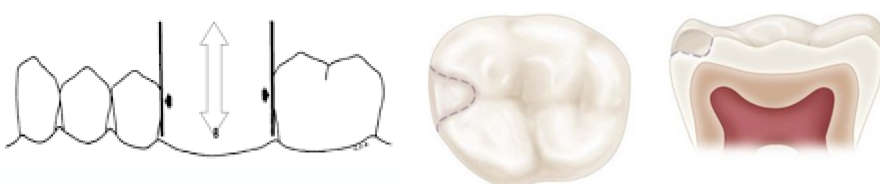
6. **Prosthetic Mouth Preparation** (for definitive RPD)

It includes preparing rest seats and guide planes on the abutment teeth.

The rest seat is a depression created usually on the occlusal surface of the teeth to receive a rest.

A rest is a metallic extension of the partial dentures created to transfer occlusal load to the abutment teeth.

Guide planes are parallel surfaces created on the proximal surface of abutment teeth along which certain parts of the partial denture will slide across during insertion and removal. The guide planes help to provide tensile-frictional resistance/indirect retention to the proximal plate of the partial denture.



(pocket dentistry)

7. **Secondary or Final Impression Making and Master Cast**

The secondary impression is made after prosthetic mouth preparation. Polyvinyl silicone impression materials are usually used. The master cast is poured using dental stone. Stone is preferred due to its high strength and surface reproduction. The master cast is surveyed, and the markings made on the primary cast are transferred to the master cast. The master cast is used to fabricate the prosthesis.

8. **Fabricating the Framework** (definitive RPD)

The metallic framework is fabricated by casting the wax pattern. Casting is done using a refractory cast. A refractory cast is prepared by duplicating the master cast. Undercuts and relief areas in the master cast should be blocked out before duplication. The wax pattern is prepared on the refractory cast as per the planned design. The refractory cast is invested, the wax is burnt out, and then the resulting mould is cast using molten metal. The material to be used to fabricate the framework is determined during the treatment planning phase. After casting, the framework should be finished and polished.

9. **Fabricating the occlusal rims**

The occlusal rims are fabricated over the temporary denture base. The jaw relationships are recorded. A face-bow and a fully adjustable articulator should be used for cases requiring high precision, like full mouth rehabilitation (occlusal reconstruction for patients with severe attrition or tipped teeth with loss of a proper plane of occlusion, where the rehabilitated occlusion should be perfect) and for removable partial denture opposing fixed partial denture, etc. Usually, jaw relation and articulation (mounting) are done the same way as a complete denture.

Teeth selection and arrangement are done on the occlusal rims.

10. Try-in of the Trial Denture

The trial denture is placed in the patient’s mouth for aesthetic and functional evaluation before processing. Error correction and modifications according to the requirements of the patient are completed.

11. Processing the Trial Denture

After try-in, wax up, flasking, de-waxing, packing, curing, finishing and polishing procedures are carried out.

12. Denture Insertion

The denture is inserted and evaluated. Premature occlusal contacts and minor errors are corrected. Post-insertion instructions are given to the patient. The patient is recalled frequently to evaluate the tissue response, obtain feedback and determine periodically the success of the denture.

Retention: Resistance to removal from the tissues or teeth

Stability: Resistance to movement in a horizontal direction (antero-posteriorly or medio-laterally)

Support: Resistance to movement towards the tissues or teeth

Abutment: A tooth that supports a partial denture.

References:

Singh, M.P., Anwar, M., & Kumar, L. (2016). Shortened dental arch. *Dental Abstracts*.

Mijiritsky, E., Lorean, A., Mazor, Z., & Levin, L. (2015). Implant Tooth-Supported Removable Partial Denture with at Least 15-Year Long-Term Follow-Up. *Clinical implant dentistry and related research, 17 5*, 917-22 .

Virard, F., Venet, L., Richert, R. *et al*. Manufacturing of an immediate removable partial denture with an intraoral scanner and CAD-CAM technology: a case report. *BMC Oral Health* 18, 120 (2018). https://doi.org/10.1186/s12903-018-0578-3